

COS Deep-Dive — Priority Question Set

Computational Finance (WI4151), TU Delft — Prof. Fang Fang

Contents

1	Mastery checklist (tick when you can do it cold)	1
2	High-probability questions (have Claude Code grill me on these)	1
3	My answers (concise board versions — re-derive these out loud, don't read)	5
4	A. Density recovery	6
5	B. Why cosine	6
6	C. Pricing formula	6
7	D. Payoff coefficients V_k	7
8	E. Truncation range (THE bug)	7
9	F. Convergence	8
10	G. Greeks	8
11	H. The pricing problem	8
12	I. Reflection principle (derive)	9
13	J. PDE routes	9
14	K. COS for discrete barriers (the main event)	9

Fang invented COS. If she asks one hard question, it is probably here. Goal: I can do everything on the checklist **closed-book, on paper, out loud**.

1 Mastery checklist (tick when you can do it cold)

- ☐ Derive the COS pricing formula for a European option from the Fourier-cosine expansion of the density.
- ☐ State where the characteristic function enters and why COS only needs the cf, not the density.
- ☐ Explain why a **cosine** expansion (even extension, real coefficients) rather than a full complex Fourier series.
- ☐ Write the payoff coefficients V_k for a call/put and explain the closed form.

- Choose the truncation range $[a, b]$ from cumulants: $a, b = c_1 \pm L \cdot \sqrt{c_2 + \sqrt{c_4}}$. Explain each term.
- Explain the convergence rate: exponential for smooth densities, algebraic otherwise — and what kills it.
- **My Assignment 2 bug:** why a small c_2 (short maturity) makes $[a, b]$ too narrow to cover the spot grid, and exactly how I fixed it.
- COS for Heston: plug in the Heston cf; state the Feller condition.
- COS for Barrier options (Lecture 10) — how the method adapts.
- COS for American options (Lecture 9) — the approach used.
- COS vs Carr–Madan FFT: when and why COS wins.

2 High-probability questions (have Claude Code grill me on these)

1. Derive the COS European call price. Where does the cf enter?

A. Start from the discounted-expectation price in the log-state y (e.g. $y = \ln(S_T/K)$):

$$P(x_0, T) = e^{-rT} \int_{-R}^R V(y, T) f(y|x_0) dy.$$

I do not know f , but I know its characteristic function $\varphi_{\{X_T\}}(t; x_0) = E[e^{it X_T} | x_0] = \int e^{ity} f(y) dy$, i.e. φ is the Fourier transform of f .

Step 1 — truncate. Pick $[a, b]$ (cumulant rule, Q2) containing essentially all the mass and restrict the integral to $[a, b]$.

Step 2 — cosine-expand the density. On $[a, b]$ write the half-range cosine series (prime halves $k = 0$):

$$f(y) \approx \sum'_{k \geq 0} A_k \cos(k\pi (y-a)/(b-a)), \quad A_k = (2/(b-a)) \int_a^b f(y) \cos(k\pi (y-a)/(b-a)) dy$$

Step 3 — A_k from φ (the engine). Write $\cos = \text{Reexp}$, pull out the phase $e^{-ik\pi a/(b-a)}$, and extend the integral to R (the *only* approximation — range truncation):

$$A_k \approx F_k = (2/(b-a)) \text{Re} \{ \varphi(k\pi/(b-a); x_0) \cdot \exp(-ik\pi a/(b-a)) \}.$$

The remaining $\int_{-R}^R f(y) e^{i(k\pi/(b-a))y} dy = \varphi(k\pi/(b-a))$ exactly. No integral survives.

Step 4 — payoff coefficients. Define $V_k = (2/(b-a)) \int_a^b V(y, T) \cos(k\pi (y-a)/(b-a)) dy$. For the call $V = K(e^y - 1)^+ \text{ lives on } y \geq 0$, so

$$V_k^{\text{call}} = (2/(b-a)) K [\chi_k(0, b) - \psi_k(0, b)].$$

Step 5 — assemble. Insert the expansion, swap $\Sigma/\int$ (uniform convergence of the cosine series for smooth f), and use V_k :

$$P \approx e^{-rT} (b-a)/2 \sum'_{k=0}^{N-1} A_k V_k.$$

Since $(b-a)/2 \cdot A_k = \text{Re} \{ \varphi(k\pi/(b-a); x_0) e^{-ik\pi a/(b-a)} \}$, this is

$$P \approx e^{-rT} \sum'_{k=0}^{N-1} \text{Re} \{ \varphi(k\pi/(b-a); x_0) \exp(-ik\pi a/(b-a)) \} V_k.$$

Where the cf enters: in exactly one place — as the cosine coefficients of the density, $\text{Re} \{ \varphi \cdot e^{-ik\pi a/(b-a)} \}$. The spot dependence sits inside $\varphi(\cdot; x_0)$; for BS/Lévy $\varphi(u; x_0) = \varphi_{\text{BS}}(u) e^{iux_0}$, so one set of V_k prices the whole spot/strike grid (only the cheap e^{iux_0} changes). Cost $O(N)$, no integration, no FFT.

2. How do you pick $[a, b]$? What breaks at short maturity, and how did you fix it in your assignment?

A. Cumulant rule (Fang–Oosterlee):

$$[a, b] = [c_1 - L\sqrt{c_2 + \sqrt{c_4}}, c_1 + L\sqrt{c_2 + \sqrt{c_4}}], \quad L \in [8, 12] \text{ (typ. 10)},$$

where c_1 = mean (centres the box), c_2 = variance ($\sqrt{c_2}$ is the natural width scale, so $L\sqrt{c_2}$ is an “L-sigma” box), $\sqrt{c_4}$ (fourth cumulant) fattens the box for heavy tails / kurtosis (zero for Gaussian), and L is the safety multiplier — larger L lowers the truncation floor but needs more terms.

BS cumulants of $\ln(S_T/K)$:

$$c_1 = \ln(S_0/K) + (r - \frac{1}{2}\sigma^2)T, \quad c_2 = \sigma^2 T, \quad c_3 = c_4 = 0 \quad \Rightarrow \quad [a, b] = c_1 \pm L\sigma\sqrt{T}.$$

What breaks as $T \rightarrow 0$: $c_2 = \sigma^2 T \rightarrow 0$, so the half-width $L\sqrt{c_2 + \sqrt{c_4}} \rightarrow 0$ and the box collapses toward the single point c_1 . With $c_4 = 0$ (Gaussian) there is no $\sqrt{c_4}$ floor to keep it open. Two failure modes: (i) the box gets too narrow to cover where $V \cdot f$ has mass; (ii) when pricing a whole grid of spots/strikes, a range centred by c_1 for one (S_0, K) is centred away from other grid points, so the cosine reconstruction is evaluated where it is meaningless — large, erratic errors at short maturity.

Diagnostic signature (this is the tell): the error *grows* as $T \rightarrow 0$ and does *not* improve when N increases. That distinguishes a *range* error (a fixed accuracy floor) from a *series* error (curable by more terms).

Assignment-2 fix: the truncation range was too narrow at short maturities because c_2 was tiny. I (i) raised L at short T , (ii) imposed a minimum half-width floor so $b - a$ cannot collapse (equivalently kept a $\sqrt{c_4}$ -type term), and (iii) made sure the range was computed to cover the entire spot/strike grid actually being priced, not a single point. After that the error stopped growing as $T \rightarrow 0$ and reverted to exponential decay in N .

3. Why cosine series instead of a complex Fourier series?

A. Three reasons:

- **(i) Real coefficients / real arithmetic.** The even extension of f on $[a, b]$ kills every sine term, so the surviving A_k are real (F_k is literally a *Re*·). A full complex series carries twice the coefficients and complex bookkeeping for no gain.
- **(ii) Direct sampling from φ .** The cosine coefficient *is* the real part of φ evaluated at the grid frequency $k\pi/(b-a)$: $A_k = (2/(b-a)) \operatorname{Re}\{\varphi(k\pi/(b-a)) e^{-ik\pi a/(b-a)}\}$. The cosine basis is exactly the one whose coefficients are read off the cf in one line — no inversion integral.
- **(iii) Payoff side closes in elementary functions.** V_k are integrals of (polynomial/exponential) $\times \cos$, which integrate to the closed forms χ_k, ψ_k . With a complex-exponential basis the call coefficients are not as clean. The whole method pairs the cosine coefficients of f (from φ) against those of V (closed form).

4. What sets the convergence rate? Give a case where it is only algebraic.

A. Three error sources: (1) **range truncation** $[a, b]$ — *fixed* once chosen, it sets the accuracy *floor* at which convergence saturates (exponentially small in the half-width for exponentially-decaying tails); (2) **series truncation** at N — governed by the decay of the cosine coefficients A_k ; (3) **cf/model error** — feeds straight through.

The rate in N is set by the **smoothness of f** :

- f analytic in a strip / $f \in C^\infty$ with integrable derivatives $\Rightarrow A_k$ decay **exponentially** $\Rightarrow$ **exponential** convergence in N (machine precision in dozens–low hundreds of terms).
- f with only β integrable derivatives $\Rightarrow A_k = O(k^{-(\beta+1)}) \Rightarrow$ only **algebraic** convergence.
Lack of smoothness is what kills the exponential rate.

Algebraic case: a low-regularity density — e.g. Variance-Gamma with small ν (the density develops a non-smooth peak / cusp), or a short-dated density approaching a Dirac, or near a jump/kink. There A_k decay only polynomially and you need many more terms.

5. Write the Heston characteristic function you used as ground truth. State the Feller condition.

A. Heston: $dS = rSdt + \sqrt{v}SdW1$, $dv = \kappa(\bar{v} - v)dt + \eta\sqrt{v}dW2$, $d\langle W1, W2 \rangle = \rho dt$, with κ mean-reversion speed, $\bar{v}$ long-run variance, η vol-of-vol, ρ correlation, $v0$ initial variance. The characteristic function of the log-price $\mathbf{x_T} = \ln(S_T/S0)$ (or $\ln S_T$) is affine-exponential:

$$\begin{aligned} \varphi(\mathbf{u}; T) = & \exp(i \mathbf{u} (\ln S0 + rT) \\ & + (v0/\eta^2) \cdot ((1 - e^{-DT})/(1 - G e^{-DT})) \cdot (\kappa - i\rho\eta\mathbf{u} - D) \\ & + (\kappa \bar{v}/\eta^2) \cdot ((\kappa - i\rho\eta\mathbf{u} - D)T - 2 \ln((1 - G e^{-DT})/(1 - G)))), \end{aligned}$$

with

$$\begin{aligned} D &= \sqrt{(\kappa - i\rho\eta\mathbf{u})^2 + (\mathbf{u}^2 + i\mathbf{u}) \eta^2}, \\ G &= (\kappa - i\rho\eta\mathbf{u} - D) / (\kappa - i\rho\eta\mathbf{u} + D). \end{aligned}$$

D and G come from solving the Riccati ODE for the variance factor of the affine system. (Equivalent “little-trap” form uses G rather than $\mathbf{g}=1/G$ to keep the complex log on the principal branch and avoid branch-cut discontinuities — the cf must be evaluated continuously or COS picks up φ -error that feeds straight through.)

Feller condition: $2\kappa\bar{v} \geq \eta^2$. It guarantees the CIR variance process stays strictly positive (never hits 0), which keeps the density well-behaved and φ clean; if violated, v can touch zero and the simulation/quadrature near $v = 0$ needs care.

6. How does COS extend to Barrier options? To American options?

A. Barrier (Lecture 10), discretely monitored on $\mathbf{t_1} < \dots < \mathbf{t_M} = T$. Knock-out survival is a product of indicators $\prod_m 1_{\{\mathbf{x_m} < h\}}$ ($h = \ln(H/K)$). By the **Markov property** the joint density factorises into one-step transitions, so the M -dimensional integral nests into a **backward recursion**. Define continuation and option value at each date:

$$\begin{aligned} c(\mathbf{x}, \mathbf{t_m-1}) &= e^{-r\Delta t} \int v(\mathbf{y}, \mathbf{t_m}) f(\mathbf{y}|\mathbf{x}) d\mathbf{y}, \\ v(\mathbf{x}, \mathbf{t_m-1}) &= R \text{ (knocked out) for } \mathbf{x} \geq h, \quad = c(\mathbf{x}, \mathbf{t_m-1}) \text{ (alive) for } \mathbf{x} < h. \end{aligned}$$

Each continuation integral is *exactly* the European COS sum:

$$\hat{c}(\mathbf{x}, \mathbf{t_m-1}) = e^{-r\Delta t} \sum_k F_k(\mathbf{x}) V_k(\mathbf{t_m}), \quad F_k(\mathbf{x}) = \text{Re}\{\varphi(k\pi/(b-a); \mathbf{x}) e^{-ik\pi a/(b-a)}\}.$$

The new ingredient: the “payoff” $v(\mathbf{y}, t_m)$ is produced by the previous step, so its coefficients V_k are **propagated by backward induction**. Because v is piecewise at h , V_k splits at the barrier:

$$\hat{V}_k(\mathbf{t_m}) = \hat{C}_k(a, h, \mathbf{t_m}) + e^{-r(T-\mathbf{t_m-1})} (2R/(b-a)) \Psi_k(h, b),$$

and $\hat{C}_k$ (inserting the COS continuation) becomes $\sum_j \varphi_{1v1}(j\pi/(b-a)) V_j M_{\{k, j\}}$ with an analytic coupling $M_{k, j}$. $M_{k, j}$ splits into a **Hankel** part ($j + k$) and a **Toeplitz** part ($j - k$); Toeplitz/Hankel $\times$ vector = convolution = 3 FFTs by circulant embedding $\Rightarrow O(N \log N)$ per step, total $O((M-1)N \log_2 N)$. So — unlike European COS — the barrier method *does* use the FFT, because it must move the whole coefficient vector at every step. “In” barriers come from in–out

parity. Continuous monitoring is the $M \rightarrow \infty$ limit, recovered by Richardson extrapolation in Δt using the known $O(\sqrt{\Delta t})$ discrete-vs-continuous bias. Heston $\Rightarrow$ 2-D state (log-stock, variance): 1-D COS in log-stock + numerical integration over variance, or full 2-D COS; Feller $2\kappa\bar{v} \geq \eta^2$.

B. American (Lecture 9). Same backward-recursion idea but the decision is *early exercise* rather than a barrier knock-out. On a time grid, at each step the option value is $v(\mathbf{x}, \mathbf{t}_m) = \max(g(\mathbf{x}), c(\mathbf{x}, \mathbf{t}_m))$ — the larger of immediate exercise payoff g and the COS continuation value $c(\mathbf{x}, \mathbf{t}_m) = e^{-r\Delta t} \sum_k F_k(\mathbf{x}) V_k(\mathbf{t}_{m+1})$. The early-exercise boundary x^*_m is the point where $g = c$; below/above it (call vs put) the option is exercised. So each step (i) finds x^*_m (a 1-D root-find on $g - c = 0$), (ii) splits the cosine coefficients of v at x^*_m into a continuation region (cosine coeffs of c , computed from the previous V_k via the same analytic coupling integrals) and an exercise region (cosine coeffs of g , closed-form χ_k/ψ_k). This is the COS Bermudan pricer; the American price is obtained by Richardson extrapolation in the number of exercise dates. Structurally identical to the barrier recursion, with the moving boundary x^*_m playing the role the fixed barrier h plays there.

7. Compare COS with Carr–Madan. Advantages and limitations.

A. Both need only the characteristic function. Carr–Madan damps the call ($c_T(\mathbf{k}) = e^{-\alpha \mathbf{k}} C_T(\mathbf{k})$, $\alpha > 0$, to make it $L^1 \cap L^2$), Fourier-transforms it in closed form in terms of φ , and recovers the price by an **inverse FFT** — i.e. a trapezoidal-type quadrature of an oscillatory integrand, $O(N \log N)$.

	COS	Carr–Madan / FFT
Core op	one $O(N)$ cosine sum, no integration	FFT quadrature, $O(N \log N)$
Free params	none (range from cumulants)	damping α must be tuned
Convergence (smooth f)	exponential in N	algebraic (quadrature-limited)
Terms for ~machine prec.	dozens–hundreds	thousands of nodes
Greeks	analytic, same sum	extra transforms / finite differences

COS wins for smooth/piecewise-smooth densities with known φ and exponentially-decaying tails — essentially all European pricing under BS, Heston, and standard Lévy/AJD models — because it converges exponentially, needs no free parameter, and gives Greeks for free.

Limitations / be careful: genuinely low-regularity densities (slow algebraic A_k decay), very short maturities, or fat tails, where $[a, b]$ must be chosen carefully (Q2) and N raised; and any error in φ (e.g. a mis-evaluated Heston cf across the branch cut) feeds straight through. Carr–Madan’s FFT also naturally returns a *whole strip of strikes* in one transform, which COS matches only because its V_k are strike-independent in log-moneyness.

3 My answers (concise board versions — re-derive these out loud, don’t read)

Compressed versions of the full answers above; expand any of them on demand.

- COS call price.** $P = e^{-rT} \int V f dy$; I don’t know f but know its cf φ . Truncate to $[a, b]$, cosine-expand f , and the coefficients come straight off φ : $A_k \approx (2/(b-a)) \operatorname{Re}\{\varphi(k\pi/(b-a)) e^{-ik\pi a/(b-a)}\}$ (extend the coefficient integral to R — the only approximation). Pair against payoff coeffs

$V_k = (2/(b-a)) \int_a^b V \cos(\dots) dy$; the integral of the product is $(b-a)/2 \sum A_k V_k$, giving $P \approx e^{-rT} \sum_{k=0}^{N-1} \text{Re}\{\varphi(k\pi/(b-a); x_0) e^{-ik\pi a/(b-a)}\} V_k$. The cf enters only as the cosine coefficients of the density; x_0 lives inside φ , so one V_k prices the whole grid.

2. $[a, b]$. $c_1 \pm L\sqrt{c_2 + \sqrt{c_4}}$, $L \approx 10$: c_1 centres, $\sqrt{c_2}$ is the width scale, $\sqrt{c_4}$ fattens for fat tails, L is the safety margin. BS: $c_1 = \ln(S_0/K) + (r - \frac{1}{2}\sigma^2)T$, $c_2 = \sigma^2 T$, $c_4 = 0$, so $[a, b] = c_1 \pm L\sigma\sqrt{T}$. As $T \rightarrow 0$, $c_2 \rightarrow 0$ so the box collapses (no $\sqrt{c_4}$ floor in the Gaussian case) $\rightarrow$ error *grows* as $T \rightarrow 0$ and is *not* cured by more N (that signature = range, not series, error). Assignment-2 fix: raise L at short T , impose a minimum half-width floor, and size the range to the whole spot/strike grid, not one point.
3. **Why cosine.** Even extension $\Rightarrow$ real A_k , real arithmetic, half the storage; A_k is literally $\text{Re}\varphi$ at the grid frequency (direct cf sampling); payoff coeffs reduce to elementary χ_k , ψ_k . A complex series gives none of these cleanly.
4. **Convergence.** Three errors: range (fixed floor), series (decay of A_k), cf/model (passes through). Rate set by smoothness of f : analytic $\Rightarrow$ exponential A_k decay $\Rightarrow$ exponential in N ; only β integrable derivatives $\Rightarrow A_k = O(k^{-(\beta+1)}) \Rightarrow$ algebraic. Algebraic case: VG with small ν , or short-dated density near a Dirac/kink.
5. **Heston cf + Feller.** Affine-exponential cf in u with $D = \sqrt{((\kappa - i\rho\eta u)^2 + (u^2 + iu)\eta^2)}$, $G = (\kappa - i\rho\eta u - D)/(\kappa - i\rho\eta u + D)$, variance term $(v_0/\eta^2)((1 - e^{-DT})/(1 - Ge^{-DT}))(\kappa - i\rho\eta u - D)$ plus the $(\kappa\bar{v}/\eta^2)$ drift term; use the “little-trap” form for a continuous branch. Feller: $2\kappa\bar{v} \geq \eta^2$ keeps $v > 0$ and φ clean.
6. **Barrier / American.** Both = backward recursion via Markov factorisation; each step is European $\text{COS } \hat{c} = e^{-r\Delta t} \sum F_k(x) V_k(t_m)$ with V_k propagated. Barrier: knock out by overwriting $v = R$ above h , V_k splits at h , the coupling $M_{k,j}$ is Hankel+Toeplitz $\Rightarrow$ FFT, $O((M-1)N \log_2 N)$. American: $v = \max(g, c)$, split V_k at the early-exercise boundary x^*_m (root of $g - c$); Bermudan + Richardson in #exercise dates. The moving boundary x^* plays the role of the fixed barrier h .
7. **COS vs Carr–Madan.** Both need only φ . CM damps the call (param α) and inverts by FFT quadrature — algebraic, $O(N \log N)$, tunable α . COS is a single $O(N)$ cosine sum, no integration, no free parameter, exponential convergence, free Greeks. COS wins for smooth densities with decaying tails; be careful for low-regularity / very short T / fat tails (set $[a, b]$, raise N), and any φ error feeds straight through.

Lecture 6 drill set (European COS)

Companion to [notes/Lecture06-notes.pdf](#). Each question has a terse answer key in the collapsible note — cover it and answer out loud first. “Derive” means on paper, no notes.

4 A. Density recovery

A1. Write the half-range cosine expansion of a density f on $[a, b]$, with the prime convention. What does the prime mean and why is $A_0/2$ the mean of f over $[a, b]$? > Key: $f(x) = \sum A_k \cos(k\pi(x-a)/(b-a))$, $A_k = (2/(b-a)) \int_a^b f \cos(k\pi(x-a)/(b-a)) dx$. Prime halves $k=0$. $A_0 = (2/(b-a)) \int f = (2/(b-a)) \cdot (\text{mass})$, and $A_0/2$ over the constant basis function is the average height.

A2. (*Core derivation.*) Derive $A_k \approx (2/(b-a)) \text{Re}\{\varphi(k\pi/(b-a)) \exp(-ik\pi a/(b-a))\}$ starting from the coefficient integral. Identify the single approximation you make and where. > Key: $\cos = \text{Re of exp}$; pull

out $e^{-ik\pi a/(b-a)}$; the remaining $\int_a^b f e^{i(k\pi/(b-a))x} dx$ is extended to $\mathbb{R}$ (THE approximation — range truncation) $= \varphi(k\pi/(b-a))$. Everything else exact.

A3. Standard normal: $\varphi(t)=e^{-t^2/2}$. Write F_k for $a=-b$ symmetric. What range do the slides suggest and why (10^{-8} clip)? > Key: substitute $\varphi(t)=e^{-t^2/2}$ at $t=k\pi/(b-a)$ into F_k , giving $F_k=(2/(b-a))\text{Re}\{e^{-\frac{1}{2}(k\pi/(b-a))^2} e^{-ik\pi a/(b-a)}\}$. The slides set $a=\Phi^{-1}(1e-8)$, $b=-a$, i.e. clip each tail at probability $1e-8$ — symmetric because the standard normal is symmetric about 0. The Gaussian factor $e^{-\frac{1}{2}(k\pi/(b-a))^2}$ makes F_k decay super-exponentially in k , which is the canonical exponential-convergence picture.

5 B. Why cosine

B1. Three reasons COS uses a cosine series, not a complex Fourier series. Be specific about each. > Key: (i) even extension $\Rightarrow$ real coefficients, real arithmetic, half the storage; (ii) A_k is literally $\text{Re}\{\varphi$ at grid frequency $\}$ — direct sampling from ch.f.; (iii) payoff coefficients V_k reduce to elementary χ_k , ψ_k (exp/poly $\times$ cos integrate in closed form).

6 C. Pricing formula

C1. (*Core derivation.*) From $P = e^{-rT} \int V f dy$, derive $P \approx e^{-rT} \sum_{k=0}^{N-1} \text{Re}\{\varphi(k\pi/(b-a); x_0) e^{-ik\pi a/(b-a)}\} V_k$. State the V_k definition and the sum–integral swap justification. > Key: insert cosine expansion of f , swap $\Sigma/\int$ (uniform convergence, smooth f), define $V_k=(2/(b-a)) \int_a^b V \cos(\dots) dy \Rightarrow P \approx e^{-rT} (b-a)/2 \sum A_k V_k$; sub $(b-a)/2 \cdot A_k = \text{Re}\{\varphi e^{-ik\pi a/(b-a)}\}$; truncate at N .

C2. Where exactly does the ch.f. enter, and where does the spot x_0 dependence live? Why does one set of $\{V_k\}$ price the whole strike/spot grid? > Key: φ enters only as the cosine coefficients of the density, $\text{Re}\{\varphi e^{-ik\pi a/(b-a)}\}$ — nowhere else. The spot x_0 sits inside $\varphi(\cdot; x_0)$; for Lévy/BS $\varphi(u; x_0) = \varphi_{\text{lvl}}(u) e^{iux_0}$, so across a strike/spot grid only the cheap e^{iux_0} phase changes while the level of φ_{lvl} is reused. The payoff coeffs V_k are strike-independent in the log-moneyness variable, so they are computed once and reused for the whole grid — that is the source of COS’s grid efficiency.

C3. What is the per-price computational cost of the COS sum? Does it call an FFT? > Key: $O(N)$ per price — a single finite cosine sum of N terms, no numerical integration and no FFT. Each term is one cf evaluation times a precomputed V_k , so the whole price is essentially a dot product. This is the headline contrast with Carr–Madan ($O(N \log N)$ FFT quadrature) and with the discrete-barrier COS (which does need the FFT because it propagates the full coefficient vector at every step).

7 D. Payoff coefficients V_k

D1. (*Derive.*) Define $\chi_k(c,d) = \int_c^d e^y \cos(\omega_k(y-a)) dy$, $\omega_k = k\pi/(b-a)$. Derive the closed form. > Key: $\chi_k = \text{Re}\{e^{-i\omega_k a} / (1+i\omega_k) [e^{\{(1+i\omega_k)y\}}]_c^d\}$; rationalise by $(1-i\omega_k)/(1+\omega_k^2)$; $\chi_k = 1/(1+\omega_k^2) [\cos(\omega_k(d-a))e^d - \cos(\omega_k(c-a))e^c + \omega_k \sin(\omega_k(d-a))e^d - \omega_k \sin(\omega_k(c-a))e^c]$.

D2. Write $\psi_k(c,d)$ including the $k=0$ case. Why must $k=0$ be special-cased? > Key: for $k \neq 0$, $\psi_k(c,d) = (1/\omega_k) [\sin(\omega_k(d-a)) - \sin(\omega_k(c-a))] = (b-a)/(k\pi) [\sin(\omega_k(d-a)) - \sin(\omega_k(c-a))]$. For $k=0$, $\omega_0=0$ so the integrand $\cos(0)=1$ and $\psi_0=d-c$. The $k=0$ case must be special-cased because the general $1/\omega_k$ formula is $0/0$ there (a removable singularity); in code a naive division by $\omega_0=0$ gives NaN, so the $k=0$ term — which also carries the $\frac{1}{2}$ from the prime — is set explicitly.

D3. In $y=\ln(S_T/K)$, give V_k for call and put. Why is the call interval $[0,b]$ and the put interval $[a,0]$?
 > Key: call $V_k=(2/(b-a))K[\chi_k(0,b)-\psi_k(0,b)]$ — payoff $K(e^{y-1})$ + lives on $y\geq 0$; put $V_k=(2/(b-a))K[\psi_k(a,0)-\chi_k(a,0)]$ — payoff $K(1-e^y)$ + lives on $y\leq 0$.

8 E. Truncation range (THE bug)

E1. State $[a,b]=[c_1 - L\sqrt{(c_2+\sqrt{c_4})}, c_1 + L\sqrt{(c_2+\sqrt{c_4})}]$. Interpret each of $c_1, c_2, \sqrt{c_4}, L$. > Key: c_1 mean (centres), c_2 variance ($\sqrt{c_2}$ width scale, L -sigma box), $\sqrt{c_4}$ fattens box for heavy tails/kurtosis (0 for Gaussian), L safety multiplier (8–12, typ. 10).

E2. Give the BS cumulants of $\ln(S_T/K)$ and the resulting box. Which cumulants vanish? > Key: $c_1=\ln(S_0/K)+(r-\frac{1}{2}\sigma^2)T$, $c_2=\sigma^2T$, $c_4=0$ (and $c_3=0$) $\Rightarrow [a,b]=c_1 \pm L\sigma\sqrt{T}$.

E3. (*High priority.*) As $T\rightarrow 0$, what happens to $[a,b]$ and why? Give the diagnostic signature distinguishing a range problem from a series problem. How do you fix it? > Key: $c_2=\sigma^2T\rightarrow 0 \Rightarrow$ half-width $\rightarrow 0 \Rightarrow$ box collapses to $\sim c_1$; with $c_4=0$ no floor keeps it open. Signature: error GROWS as $T\rightarrow 0$ and does NOT improve when N increases (range error is a floor, not curable by more terms). Fix: raise L at short T ; keep $\sqrt{c_4}$ / impose a minimum half-width floor; ensure the range covers the whole spot/strike grid actually priced, not one point.

9 F. Convergence

F1. List the three error sources in the COS price and say which one sets the accuracy floor. > Key: (1) range truncation $[a,b]$ — FIXED once chosen, sets the floor; (2) series truncation at N — set by decay of A_k ; (3) ch.f./model error feeds straight through. Convergence saturates at the range floor.

F2. What property of f gives exponential vs only algebraic convergence in N ? Give a case where it is only algebraic. > Key: smoothness/analyticity $\Rightarrow A_k$ decay exponentially $\Rightarrow$ exponential in N . Finite smoothness (β integrable derivatives) $\Rightarrow A_k=O(k^{-(\beta+1)}) \Rightarrow$ algebraic. Algebraic case: density with low regularity / near a kink/jump, e.g. VG with small ν , or short-dated densities approaching a Dirac.

F3. COS vs MC convergence rate? Roughly how many terms for machine precision on Heston (cf slide 31)? > Key: MC converges at $O(N^{-1/2})$ in the number of paths, whereas COS converges exponentially in the number of terms for smooth densities. On Heston (slide 31) $N\sim 160$ terms reach $\sim 1e-10$ error, versus thousands of FFT nodes for worse error and longer CPU time. So COS reaches machine precision with N in the dozens to low hundreds — the gap is the whole selling point of the method.

10 G. Greeks

G1. Why are Delta and Vega “free” in COS? Write the Delta sum. > Key: S_0 (via $x=\ln(S_0/K)$) and the model parameters enter the COS sum only analytically, through $\varphi(\cdot;x_0)=\varphi(\cdot)e^{\{iux\}}$; so differentiation passes straight through the finite sum with no re-integration. $\Delta \approx e^{-rT}\Sigma' \operatorname{Re}\{\varphi(k\pi/(b-a)) e^{\{ik\pi(x-a)/(b-a)\}} \cdot ik\pi/(b-a)\} V_k/S_0$ (chain rule $\partial x/\partial S_0=1/S_0$). Vega differentiates φ w.r.t. the variance parameter instead. The same $\{V_k\}$ are reused, so Greeks cost essentially nothing extra — a structural advantage over quadrature/FFT pricers that need finite differences.

11 H. The pricing problem

H1. Define up-and-out vs up-and-in call. Why are there exactly 8 barrier types? > Key: up-and-out call pays $\max(S_T - K, 0)$ UNLESS the asset crosses $B > S_0$ during $[0, T]$, in which case it pays 0 (knocked out); up-and-in call pays 0 UNLESS B is crossed, in which case it becomes a vanilla call (knocked in). The 8 types = up/down $\times$ in/out $\times$ call/put ($2 \times 2 \times 2$). Each has a closed-form GBM price.

H2. State in-out parity and give its two practical uses. > Key: holding knock-in and knock-out with the same (K, B, T) reproduces the vanilla, because exactly one of the two is alive at maturity: $C_{\text{in}} + C_{\text{out}} = e^{-r(T-t)} E[(S_T - K)^+]$. Use (i): price the easier leg (usually the knock-out) plus the vanilla, then back out the other by subtraction — cheaper and more stable than handling the in-condition directly. Use (ii): it gives a free correctness check on any numerical barrier price.

H3. Write the two equivalent formulations (expectation and PDE). Where does the barrier enter the PDE? > Key: $V = e^{-r(T-t)} E_x[(e^{(X_T - K)}) + 1_{\{\tau_B > T\}}]$; or localized Feynman–Kac BS PDE on $[a, b]$ with terminal payoff and ABSORBING boundary $V(a, t) = V(b, t) = 0$. Barrier enters via the homogeneous Dirichlet BC, not the equation.

12 I. Reflection principle (derive)

I1. (Core.) Derive $P(\tau_a \leq T) = 2P(W_T \geq a)$ and hence the density of the running maximum Ξ . > Key: split $P(\tau_a \leq T)$ by $\{W_T > a\}, \{W_T \leq a\}$; reflection $\Rightarrow$ both halves equal $\Rightarrow 2P(\tau_a \leq T, W_T \geq a) = 2P(W_T \geq a)$ (since $\{W_T \geq a\} \subseteq \{\tau_a \leq T\}$). So $\Xi \sim |W_T|$, $P(\Xi \leq a) = \text{erf-type}$, $p_\Xi(a) = \sqrt{(2/\pi T)} e^{-a^2/2T} 1_{\{a \geq 0\}}$.

I2. (Core.) Derive the joint density of (Ξ_0^T, W_T) . > Key: for $b \leq a$, reflect after τ_a : $P(\Xi \geq a, W_T < b) = P(\Xi \geq a, W_T > 2a - b) = P(W_T > 2a - b)$ (since $2a - b \geq a$). Differentiate $-\partial^2/\partial a \partial b \Rightarrow p(a, b) = \sqrt{(2/\pi T)} (2a - b)^{-3/2} e^{-(2a - b)^2/2T} 1_{\{a \geq \max(b, 0)\}}$.

I3. How do you get from standard BM to drifted BM (GBM log-price) results? > Key: Girsanov change of measure. A drifted BM $W_T + \theta t$ under P is a driftless BM under an equivalent measure Q , with Radon–Nikodym derivative $dP/dQ = e^{\theta W_T - \frac{1}{2}\theta^2 T}$. So I compute the reflection-principle result for driftless BM under Q , then reweight by that exponential to recover the drifted (GBM log-price) law. This is exactly what turns the driftless joint density of (Ξ, W_T) into the drifted one used in the closed-form barrier prices.

13 J. PDE routes

J1. State the method-of-images identity and the down-and-out call it produces. What is α ? > Key: for the BS operator $LV = V_t + rSV_S + \frac{1}{2}\sigma^2 S^2 V_{SS} - rV$, if $LV = 0$ then the image $U(S, t) = S^{2\alpha} V(B^2/S, t)$ also solves $LU = 0$, with $\alpha = \frac{1}{2}(1 - 2r/\sigma^2)$. This is the BS analogue of reflecting a heat-equation solution across a wall. Subtracting the image with the right weight cancels the value on the barrier $S = B$, giving the down-and-out call $C_{\text{down-out}} = C_{\text{van}}(S, K) - (S/B)^{2\alpha} C_{\text{van}}(B^2/S, K)$: the first term is the vanilla, the second removes exactly the paths that would have crossed B , evaluated at the mirror spot B^2/S .

J2. (Derive.) Sine-series PDE route: what change of variables reduces the localized BS PDE to the heat equation, and why does the sine basis appear? > Key: $\tau = \frac{1}{2}\sigma^2(T - t)$, $\kappa = 2r/\sigma^2$, $V = e^{\alpha x + \beta \tau} U$

with $\alpha = -\frac{1}{2}(\kappa - 1)$, $\beta = -\alpha^2 - \kappa$, $z = x - a \Rightarrow U_\tau = U_{zz}$ on $[0, b-a]$, $U(0, \tau) = U(b-a, \tau) = 0$. Homogeneous Dirichlet BCs $\Rightarrow$ sine series. Propagate modes by $e^{\{-n\pi/(b-a)\}^2 \tau}$.

14 K. COS for discrete barriers (the main event)

K1. (*High priority.*) Why does a discretely-monitored barrier become a BACKWARD RECURSION? Where does the Markov property enter? > Key: survival = $\prod \{m\} 1\{x_m < h\}$; Markov $\Rightarrow$ joint density factorises into one-step transitions $\Rightarrow$ the M-dim integral nests $\Rightarrow$ backward recursion: $c(x, t_{m-1}) = e^{\{-r\Delta t\}} \int v(y, t_m) f(y|x) dy$, then knock out: $v = R$ for $x \geq h$, $= c$ for $x < h$.

K2. Write the COS step for the continuation value. How is it the European COS formula? > Key: $\hat{c}(x, t_{m-1}) = e^{\{-r\Delta t\}} \sum_k F_k(x) V_k(t_m)$, $F_k(x) = \text{Re}[\varphi(k\pi/(b-a); x) e^{\{-ik\pi a/(b-a)\}}]$, $V_k =$ cosine coeffs of $v(y, t_m)$. It IS Euro COS with $v(\cdot, t_m)$ playing the role of the payoff — except v 's coefficients come from the previous step.

K3. (*Core.*) Why must V_k be propagated by backward induction, and how does the knock-out split V_k at h ? > Key: v is piecewise (R above h , $\hat{c}$ below) $\Rightarrow \hat{V}_k(t_m) = \hat{C}_k(a, h, t_m) + e^{\{-r(T-t_{m-1})\}} (2R/(b-a)) \Psi_k(h, b)$. $\hat{C}_k$ inserts the COS continuation value $\Rightarrow \hat{C}_k = e^{\{-r\Delta t\}} \text{Re}[\sum_j \varphi_{Levy}(j\pi/(b-a)) V_j(t_{m+1}) M_{\{k,j\}}]$, with $M_{\{k,j\}}$ an analytic coupling integral.

K4. (*High priority — contrast with Lecture 6.*) Why does the discrete-barrier COS use the FFT when European COS does not? Give the matrix structure and the complexity. > Key: $M_{\{k,j\}}$ splits into a Hankel part (depends on $j+k$) and Toeplitz part (depends on $j-k$). Toeplitz/Hankel $\times$ vector = convolution = 3 FFTs via circulant embedding $\Rightarrow O(N \log N)$ per step. Euro COS is a single $O(N)$ sum needing no coefficient propagation; the barrier needs all $\{V_k\}$ at every step. Total: $O((M-1)N \log_2 N)$.

K5. State the truncation range for the barrier problem. What extra constraint beyond Lecture 6? > Key: $[a, b] = (c1 + x0) \pm L\sqrt{(c2 + \sqrt{c4})}$, $x0 = \ln(S0/K)$, cumulants of the one-step increment. Extra: the barrier $h = \ln(H/K)$ must sit well INSIDE $[a, b]$ or the split at h is meaningless; and range is fixed once but reused for all M steps.

K6. How is Heston handled? Continuous monitoring? > Key: Heston has a 2-D state (log-stock, variance), so the one-step continuation integral is two-dimensional: either 1-D COS in the log-stock with numerical integration over the variance, or a full 2-D COS recovering the joint density. The characteristic function is Heston's (Lecture 7) and the Feller condition $2\kappa\bar{v} \geq \eta^2$ keeps it well-behaved. Continuous monitoring is the $M \rightarrow \infty$ ($\Delta t \rightarrow 0$) limit; in practice the discrete price converges to it and a Richardson-type extrapolation in Δt — using the known $O(\sqrt{\Delta t})$ discrete-vs-continuous barrier bias — recovers the continuous price.